



**SYDNEY BOYS HIGH
SCHOOL**
MOORE PARK, SURRY HILLS

2007

**YEAR 11 Mathematics 2 Unit
HALF YEARLY EXAM**

Mathematics 2 Unit

General Instructions

- Reading Time – 5 Minutes
- Working time – 90 Minutes
- Write using black or blue pen. Pencil may be used for diagrams.
- Board approved calculators may be used.
- Marks may **NOT** be awarded for messy or badly arranged work.
- All necessary working should be shown in every question.

Total Marks – 70

Examiner: *D. McQuillan*

- (1) Evaluate $\frac{5.1^2}{10.1 - 3.6}$ to 3 decimal places. [1]
- (2) Show that the triangle inequality $|x + y| \leq |x| + |y|$ is true for $x = 4$ and $y = -7$. [1]
- (3) If $\sqrt{80} = a\sqrt{5}$ find the value of a . [1]
- (4) Simplify $\frac{3a}{b} - \frac{a}{2b}$. [1]
- (5) Factorise
- (i) $x^2 + 10x - 24$ [1]
- (ii) $3m^2 - 3mn - m + n$ [1]
- (iii) $1 - 8x^3$ [1]
- (6) Rationalise the denominator of $\frac{\sqrt{5} + 7}{3\sqrt{2}}$. [1]
- (7) Find the interior angle sum of a 12 sided polygon. [1]
- (8) Write $\sqrt[4]{x^5}$ in index form. [1]
- (9) Express $0.\dot{5}\dot{4}$ as a fraction. [1]

(10) Simplify $\sqrt{5} + \sqrt{2} - \sqrt{45} + \sqrt{8}$. [2]

(11) Expand and simplify $(\sqrt{3} - 5)(\sqrt{3} + 5)$. [2]

(12) Solve these equation simultaneously [2]
 $2x - 9 = y$
 $5x + 12 = 2y$

(13) Solve [2]
(a) $|5x + 13| = |3x + 3|$.

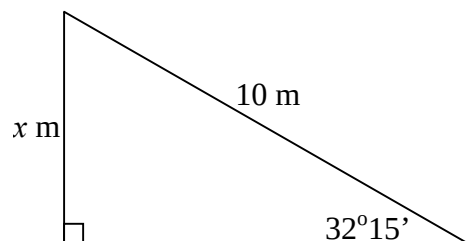
(b) $|2x - 1| > 13$. [2]

(14) Simplify $\frac{x^2 - 2x - 3}{x^2 - 4x - 5} \times \frac{x^2 - 25}{(x - 3)(x + 5)}$. [2]

(15) Use the quadratic formula to solve $2x^2 - 6x - 3 = 0$
leave your answer in simplest surd form. [2]

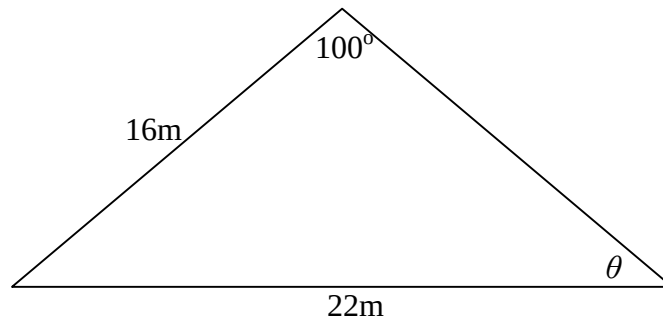
(16) Write the expression $x^2 - 6x + 12$ in the form $(x - h)^2 + k$ using the completing the square method and hence find values of h and k . [2]

(17) Find the value of x (to 2 decimal places). [2]

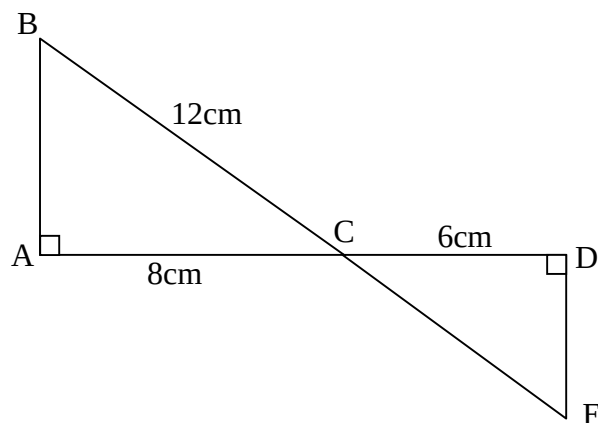


- (18) Solve
- (a) $2^{t-3} = 16$. [2]
- (b) $x^4 - 10x^2 + 9 = 0$. [2]

- (19) Find θ to the nearest degree. [2]



- (20) Given the figure



- (a) Prove that $\triangle ABC$ is similar to $\triangle DEC$. [2]
- (b) Hence find the length of CE. [1]
- (21) Find the exact value of
- (a) $\cos 45^\circ$ [1]
- (b) $\tan 180^\circ$ [1]
- (c) $\sin 300^\circ$ [2]

(22) Simplify $\cos \theta \sin(90^\circ - \theta) + \sin \theta \cos(90^\circ - \theta)$. [2]

(23) Simplify $\left(\frac{27y^6}{8}\right)^{-\frac{2}{3}}$. [2]

(24) Sketch a graph of $y = \cos x$ where $-180^\circ \leq x \leq 180^\circ$. [2]

(25) Solve $\cos \theta = -\frac{\sqrt{3}}{2}$ for $0^\circ \leq \theta \leq 360^\circ$ [2]

(26) (a) Show that $2x^2 + 9x - 5 = (2x - 1)(x + 5)$. [1]

(b) Hence solve $2x^2 + 9x - 5 \leq 0$. [1]

(c) Graph the solutions on a number line. [1]

(27) From camp, a hiker walks due north for 8 km, then 6 km due west to a lake.

(a) Draw a neat diagram to represent this information. [1]

(b) How far is the hiker from camp? [2]

(c) What is the bearing of the camp from the lake (to the nearest degree)? [2]

(28) In a right-angled triangle, one of the sides adjacent to the right angle is 4 cm longer than the other. Find the exact length of the hypotenuse if the area of the triangle is 96 cm^2 . [3]

(29) Find the points of intersection of the line $y = 3x + 2$ and the parabola $y = x^2 - 7x - 9$. [3]

(30) Show that $\frac{(\sin A + \cos A)^2}{\cos A} = 2 \sin A + \sec A$. [3]

(31) Solve $\tan^2 \phi = 1$ for $0^\circ \leq \phi \leq 360^\circ$ [3]

End of Exam

Maths 4R11 Half Yearly 2007 2 unit class.

$$\textcircled{1} \frac{26.01}{6.5} \div 4.002 \text{ (3DP)} \quad \textcircled{1} \quad \textcircled{12} \left. \begin{array}{l} y = 2x - 9 \\ 2y = 5x + 12 \end{array} \right\}$$

$$\textcircled{2} \begin{array}{l} |4 + -7| \leq |4| + |-7| \\ |-3| \leq 4 + 7 \\ 3 \leq 11 \text{ true} \end{array} \quad \textcircled{1}$$

$$\begin{array}{l} 2(2x - 9) = 5x + 12 \\ 4x - 18 = 5x + 12 \\ -30 = x \end{array}$$

$$\textcircled{3} \sqrt{80} = \sqrt{16 \times 5} = 4\sqrt{5} \Rightarrow a = 4 \quad \textcircled{1} \quad \text{and } y = 2x - 30 - 9 = -69$$

$$(-30, -69) \quad \textcircled{2}$$

$$\textcircled{4} \frac{6a - a}{2b} = \frac{5a}{2b} \quad \textcircled{1}$$

$$\textcircled{13} \text{ (a) } \begin{array}{l} 5x + 13 = 3x + 3 \\ 2x = -10 \end{array}$$

$$\textcircled{5} \begin{array}{l} \text{(i) } (x+12)(x-2) \quad \textcircled{1} \\ \text{(ii) } 3m(m-n) - 1(m-n) \\ = (3m-1)(m-n) \quad \textcircled{1} \end{array}$$

$$x = -5 \checkmark$$

$$\text{(iii) } (1-2x)(1+2x+4x^2) \quad \textcircled{1}$$

$$\begin{array}{l} 5x + 13 = -3x - 3 \\ 8x = -16 \\ x = -2 \checkmark \end{array} \quad \textcircled{2}$$

$$\textcircled{6} \frac{(\sqrt{5}+7) \times \sqrt{2}}{3\sqrt{2} \sqrt{2}} = \frac{\sqrt{10}+7\sqrt{2}}{6} \quad \textcircled{1}$$

$$\textcircled{7} \frac{360}{12} = 30 \Rightarrow \text{interior is } 150^\circ \times 12 = 1800^\circ \quad \textcircled{1}$$

$$\textcircled{8} x^{\frac{5}{4}} \quad \textcircled{1}$$

$$\begin{array}{l} \text{(b) } 2x - 1 > 13 \\ 2x > 14 \\ x > 7 \end{array}$$

$$\textcircled{9} \frac{54}{99} = \frac{6}{11} \quad \textcircled{1}$$

$$\begin{array}{l} \text{and} \\ 2x - 1 < -13 \\ 2x < -12 \\ x < -6 \end{array}$$

$$\textcircled{10} \begin{array}{l} \sqrt{5} + \sqrt{2} - 3\sqrt{5} + 2\sqrt{2} \\ = -2\sqrt{5} + 3\sqrt{2} \end{array} \quad \textcircled{2}$$

$$\textcircled{11} 3 + \cancel{5\sqrt{3}} - \cancel{5\sqrt{3}} - 25 = -22 \quad \textcircled{2}$$



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25) $\cos \theta = -\frac{\sqrt{3}}{2}$

quad 2, 3

$\cos \theta = \frac{\sqrt{3}}{2} \Rightarrow \theta = 30^\circ$

$\therefore 180 - 30 = 150^\circ$ (2)
 $180 + 30 = 210^\circ$

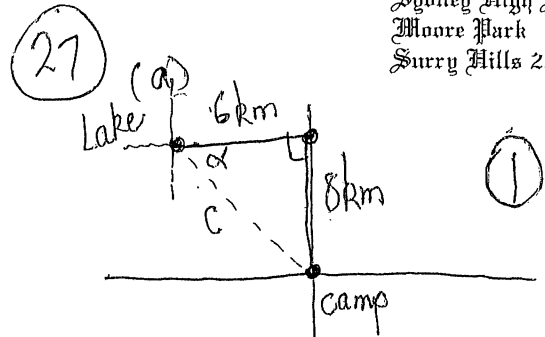
26) (a) $(2x-1)(x+5) = 2x^2 + 10x - x - 5$
 $= 2x^2 + 9x - 5$ (1)

(b) $2x^2 + 9x - 5 \leq 0$

below or touching

$-5 \leq x \leq \frac{1}{2}$ (1)

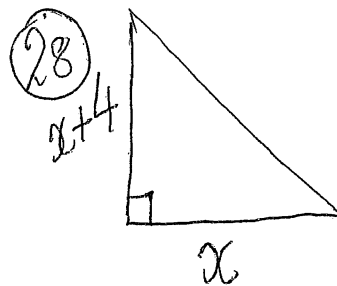
(c) (1)



(b) $6^2 + 8^2 = c^2$
 $c = 10 \text{ km}$ (2)

(c) $\sin d = \frac{8}{10}$
 $d = 53^\circ$

true bearing is $90 + 53 = 143^\circ$
or $S 37^\circ E$ (2)



area = $\frac{1}{2} \times x \times (x+4) = 96$

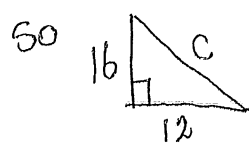
$x^2 + 4x = 192$

$x^2 + 4x - 192 = 0$

$(x+16)(x-12) = 0$

$x = -16, x = 12$

use $x = 12$



$c^2 = 12^2 + 16^2$
 $c = 20 \text{ cm}$ (3)

$$\begin{cases} (29) & y = 3x + 2 \\ & y = x^2 - 7x - 9 \end{cases}$$

$$x^2 - 7x - 9 = 3x + 2$$

$$x^2 - 7x - 3x - 9 - 2 = 0$$

$$x^2 - 10x - 11 = 0$$

$$(x - 11)(x + 1) = 0$$

$$x = -1 \text{ and } x = 11$$

$$\begin{aligned} y &= 3x - 1 + 2 \\ &= -3 + 2 \\ &= -1 \end{aligned}$$

$$(-1, -1)$$

$$\begin{aligned} y &= 3 \times 11 + 2 \\ &= 33 + 2 \\ &= 35 \end{aligned}$$

$$(11, 35)$$

(3)

$$(30) \quad \frac{\text{LHS} \quad \sin^2 A + 2 \sin A \cos A + \cos^2 A}{\cos A}$$

$$= \frac{1 + 2 \sin A \cos A}{\cos A}$$

$$= \sec A + 2 \sin A$$

$$= \text{RHS}$$

(3)

$$(31) \quad \tan^2 \phi = 1$$

$$\tan \phi = \pm 1$$

$$\tan \phi = 1$$

quad 1, 3

$$\tan \phi = -1$$

quad 2, 4

$$\phi = 45^\circ$$

$$180 - 45 = 135^\circ$$

$$180 + 45 = 225^\circ$$

$$360 - 45 = 315^\circ \quad (3)$$